

Pakistani Celebrities Gender and Name Classification

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ABSTRACT Real-world celebrity recognition is challenging because images often contain multiple faces, different lighting conditions, pose variations, occlusion, and background clutter. To address these issues, this project proposes an end-to-end two-stage deep learning pipeline for (1) face detection + gender classification and (2) celebrity identity recognition. In the first stage, a YOLOv8-based model is trained to localize faces and predict gender (Male/Female). In the second stage, the detected face regions are cropped, resized to 300×300, and normalized using ImageNet preprocessing before being passed to a fine-tuned EfficientNetB4 classifier that predicts one of 16 classes (15 Pakistani celebrity IDs + an Unknown/Other class to reduce false identification). The training pipeline uses transfer learning with full fine-tuning, data augmentation (rotation, shift, zoom, brightness, flip), AdamW optimizer, label smoothing, and callback-based control including ReduceLROnPlateau, EarlyStopping, and ModelCheckpoint. The final system produces an annotated output image containing bounding boxes and predicted labels (gender + name-class + confidence) for each detected face, enabling reliable testing on group photos and multi-person scenes. This integrated framework demonstrates how combining a strong detector (YOLO) with a dedicated classifier (EfficientNet) can improve robustness and usability in real-world celebrity face analysis tasks.

INDEX TERMS Face Detection, Celebrity Recognition, Gender Classification, YOLOv8, EfficientNetB4, Transfer Learning, Image Preprocessing, ImageNet Normalization, Data Augmentation, AdamW Optimizer, Label Smoothing, Top-K Accuracy, ModelCheckpoint, Early Stopping, Multi-Face Annotation Pipeline.

I. INTRODUCTION

Face understanding is one of the most practical problems in computer vision because faces appear everywhere social media posts, news images, TV frames, posters, and surveillance-like scenarios. However, real-world images are not “clean”: a single picture can contain multiple faces at different scales, complex backgrounds, occlusion (glasses, hair, hands), non-frontal head poses, and uneven lighting. In such conditions, a single-step “direct classification” model often fails because it first needs to find the face regions reliably before it can identify who the person is.

To address these issues this project builds a complete end-to-end pipeline that perform two task on the same Inp-imag:

- 1) **Face detection + gender prediction (Male/Female)**
- 2) **Celebrity identity recognition**, where each detected face is classified into one of 15 known Pakistani celebrities or assigned “**Unknown**” if the person is not part of the known set.

A. PROBLEM STATEMENT

MOST identity recognition errors in practical images happen before classification even starts—because the system either misses faces, crops them incorrectly, or processes background instead of face pixels. Therefore, the key challenge is to design a pipeline that:

- **Detects every face** in an image reliably, even when multiple people are present, and
- **Performs identity classification** on correctly cropped facial regions rather than on the full scene image.

This project specifically targets multi-face images, where the output must include *bounding boxes*, a *gender label*, and a *celebrity name* for every detected face.

B. MOTIVATION

In real-world images (events, interviews, posters, social media, and crowd scenes), faces appear with different scales, angles, lighting, occlusion, makeup, and backgrounds. A simple “image classification” model is not sufficient because

it assumes one dominant subject per image. Real scenes often contain multiple people and therefore require localizing faces before identity recognition.

This project is motivated by practical needs such as:

- **Media analytics:** identifying who appears in content and computing gender statistics,
- **Smart photo organization:** tagging and sorting images by identity,
- **Human-centered AI:** indexing, retrieval, and moderation support, and
- **Security and access-control prototypes:** conceptually explored, while the core focus remains recognition and research.

Moreover, introducing an **Unknown** (−1) class is essential because real-world recognition systems must handle faces that do not belong to the known identity set and avoid forced misclassification.

C. PROPOSED SOLUTION

Instead of relying on a single model to perform all tasks, the system is designed as a two-stage framework.

1) Stage 1: YOLOv8 for Face Detection and Gender Classification

The first stage uses a YOLOv8-based detector trained to localize faces and simultaneously predict gender. The gender dataset is sourced from Roboflow with a large-scale split to improve robustness across appearance variations: train (7000), valid (2000), and test (1000) images with Male/Female distributions.

To build high-quality annotations, a hybrid labeling workflow is adopted:

- **Manual annotation:** 20 images are annotated manually and then removed from the original training set to prevent duplication, and
- **Auto-annotation:** the remaining images are auto-annotated using a pre-trained YOLOv11n face-detection model.

2) Stage 2: EfficientNetB4 for Celebrity Identity Classification

Once YOLO detects a face, the detected region is cropped and resized to 300×300 , then normalized using ImageNet-style preprocessing before being passed to the classifier. The identity model is a custom Keras-based 16-class EfficientNetB4 classifier, where 15 classes represent known Pakistani celebrities and one class represents **Unknown**.

The integrated pipeline produces a final annotated image where:

- Male faces are marked with a **blue** bounding box,
- Female faces are marked with a **green** bounding box, and
- each detected face receives a predicted celebrity name (or Unknown).

This output visualization demonstrates that the model detects faces correctly and draws gender-specific bounding boxes on Male and Female faces.

II. CONTRIBUTIONS

The main contributions of this work are summarized as follows:

A. IDENTITY RECOGNITION MODEL FOR PAKISTANI CELEBRITIES

We trained and fine-tuned an EfficientNet-based classifier to recognize Pakistani celebrity identities using a folder-based dataset structure with 16 classes (15 known identities and 1 Unknown class). EfficientNet architectures are widely adopted due to their strong accuracy-to-efficiency trade-off, making them suitable for transfer learning and practical deployment.

B. TWO-STAGE INFERENCE PIPELINE (DETECTION → CLASSIFICATION)

We integrated:

- a YOLO-based model for face detection and gender prediction, and
- an EfficientNet classifier for celebrity identity recognition,

enabling the system to handle multiple faces in a single image and annotate each detected face. Ultralytics YOLO models are commonly used in real-world detection pipelines because of their speed and robustness.

C. DATASET ENGINEERING AND AUTO-ANNOTATION WORKFLOW FOR GENDER DETECTION

We built a complete workflow for gender detection training, including:

- downloading a Roboflow dataset,
- checking and handling duplicate images,
- creating a YOLO folder structure on Google Drive,
- selecting samples for manual annotation,
- auto-annotating remaining images using a pre-trained face detector, and
- training a YOLO gender detection model.

Roboflow's dataset tools and workflows are frequently used to accelerate object detection dataset preparation in applied computer vision.

D. PRACTICAL OUTPUT (READABLE LABELS AND BOUNDING BOXES)

We produced final qualitative outputs as saved annotated images containing:

- face bounding boxes,
- gender labels,
- predicted celebrity names, and
- confidence scores.



FIGURE 1. This Output Image just show Model Detect Faces Correctly and Draw bounding Boxes of Different colours on Male and Female faces.

III. COMPARATIVE LITERATURE TABLE

Here is a comparative literature table. It compares common approaches used in face detection + recognition pipelines, and shows why our approach is practical.

TABLE 1. Comparative Literature Table for Face Detection and Recognition Pipelines

Study / Method Type	Task	Typical Dataset	Model / Approach	Strengths	Limitations	Your Improvement
Classical Haar Cascades / HOG	Face detection	Small / collected	Handcrafted features	Fast on CPU, simple	Poor occlusion/angles	Deep learning detection (YOLO-based)
MTCNN-based pipelines	Detect + align face	CelebA / WIDER FACE	Cascaded CNN face detector	Good alignment, robust	Slower than modern YOLO, separate stages	Simpler two-stage pipeline, faster deployment
RetinaFace / Modern detectors	Face detection	WIDER FACE	Single-shot face detector	Very accurate for faces	Heavier models	YOLO-style approach is simpler and easier to implement
FaceNet / ArcFace (embedding-based)	Face recognition	VOGFace2, MS-Celeb	Metric learning embeddings	Scales to many identities	Needs thresholding + embedding search	Softmax classification is simpler for 15-16 classes
CNN - classification (ResNet/EfficientNet)	Identity classification	Person 10 datasets	Softmax classification	Strong for fixed identity sets	Retraining needed for new identities	Unknown = 11 classes handles "others"
Detection + classification pipelines	Detect + classify	Mixed datasets	Detect face → crop → classify	Works in real scenes	More engineering required	Full real-world multi-face pipeline implemented
Auto-annotation assisted training	Dataset creation	Custom datasets	Pre-trained detector + label export	Saves time, scalable	Risk of noisy labels	Manual + auto labeling + verification workflow

IV. METHODOLOGY

A. OVERALL SYSTEM ARCHITECTURE

Our system can be described as a three-block pipeline:

- **Block A — Identity Model Training (EfficientNet Classifier):**
 - **Input:** face images arranged by class folders
 - **Output:** trained model predicting one of 16 labels (15 known + unknown)
- **Block B — Combined Inference Pipeline (YOLO + EfficientNet):**
 - **YOLO:** Detects faces + predicts gender.
 - **Each face crop** → EfficientNet predicts identity
 - **Final Output:** Annotated image containing bounding boxes and labels in the form *Gender | Name | Confidence*.
- **Block C — Gender Detection Model Training (YOLO Training Pipeline):**
 - Dataset is downloaded from Roboflow.
 - Converted into YOLO format.
 - Manual subset selection followed by auto-annotation.
 - YOLO training and testing on test set images.

B. DATASET STRUCTURE

Distribution:

- **train/:** 7000 images (Male: 3023, Female: 3977)

- **valid/:** 2000 images (Male: 865, Female: 1135)
- **test/:** 1000 images (Male: 423, Female: 577)

All images were annotated into two classes: **Female** and **Male**.

Applied augmentations:

- Brightness/contrast
- Rotation
- Blur/noise

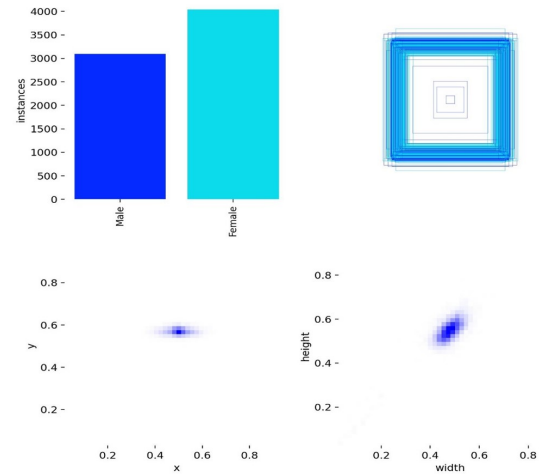


FIGURE 2. Gender dataset distribution and label statistics visualization (class counts and bounding-box distributions).

1) Identity Dataset

We assume a folder-based dataset structure as follows:

```
Data/Train/
-1/
0/
1/
...
14/
```

- **Total classes:** 16
- **Unknown class:** -1
- **Known identities:** 0-14

Runtime logs indicate approximately:

- **Training images:** 3671
- **Validation images:** 916

across 16 classes using `validation_split=0.2`.

2) Gender Dataset (Roboflow)

The gender dataset was downloaded via Roboflow and organized in the following directory structure:

```
gender-1/
train/
  Male/
  Female/
valid/
  Male/
```

```
Female/
test/
Male/
Female/
```

C. IDENTITY CLASSIFICATION MODEL (EFFICIENTNETB4)

We used EfficientNetB4 as the backbone network with the following configuration:

- **Backbone:** EfficientNetB4 with `include_top=False`, pre-trained on ImageNet
- **Pooling:** Global Average Pooling
- **Classification head:** Dense layers $512 \rightarrow 256 \rightarrow \text{Softmax}$
- **Regularization:** Dropout and Batch Normalization
- **Loss:** Categorical Cross-Entropy with label smoothing
- **Optimizer:** AdamW

EfficientNet's compound scaling strategy provides strong accuracy with good computational efficiency, making it well-suited for transfer learning on limited domain-specific datasets.

D. TRAINING STRATEGY

To improve generalization under real-world variations, we applied data augmentation during training, including:

- Rotation
- Horizontal flip
- Brightness adjustment
- Zoom
- Width/height shifts
- Shear transformation

These augmentations increase robustness to pose, illumination, and scale changes.

To stabilize training and reduce overfitting, we employed the following callbacks:

- **ReduceLROnPlateau:** decreases the learning rate when `val_loss` stops improving.
- **EarlyStopping:** stops training when validation performance plateaus.
- **ModelCheckpoint:** saves the best-performing model during training.

E. EVALUATION

Model evaluation includes:

- **Top-1 Accuracy** (main metric),
- **Top-3 Accuracy** (useful when identities are visually similar),
- **Classification Report:** including precision, recall, and F1-score per class.

This is a strong evaluation set for a multi-class identity problem.

F. MULTI-FACE INFERENCE (YOLO + IDENTITY MODEL)

The combined inference pipeline operates as follows:

- 1) Read image.
- 2) YOLO finds face bounding boxes + gender
- 3) For each detected face: crop $\rightarrow$ resize to $300 \times 300 \rightarrow$ preprocess (ImageNet normalization).
- 4) EfficientNet predicts identity class
- 5) Map class to name
- 6) Draw bounding boxes + labels + confidence

This is exactly how real multi-face recognition systems are built in applied CV. Ultralytics YOLO is commonly used for practical detection pipelines due to ease of training and deployment.

G. MODEL DEVELOPMENT WORKFLOW: ANNOTATION STEPS

Initially, I used the same dataset images used for classification. However, after I identified that there are no defined male and female classes hence, I will have to annotate all myself so, I switched to the dataset available on Roboflow and annotated into 2 steps:

1) Manual Annotation:

- 20 images were separated from train and were annotated manually on Roboflow.
- Those 20 images were then removed from train directory to avoid duplication when auto annotation.

2) Auto Annotation:

- Remaining images auto-annotated using the trained YOLOv11n (face-detection) Model.
- Link = <https://huggingface.co/AdamCodd/YOLOv11n-face-detection>

V. RESULTS

A. IDENTITY MODEL RESULTS

The identity classification pipeline produces the following quantitative outputs on the validation set:

- **Final validation Top-1 accuracy**
- **Final validation Top-3 accuracy**
- **Classification report across 16 classes** (precision, recall, and F1-score per class)

These metrics provide a reliable evaluation for the 16-class identity recognition task (15 known identities + Unknown).

B. END-TO-END QUALITATIVE RESULTS (ANNOTATED OUTPUT)

Example image is a strong demonstration:

- detects faces
- assigns correct genders
- identifies known celebrities
- assigns Unknown when identity is not recognized confidently

The system works on multi-person images, not just single-person photos.

C. GENDER DETECTION YOLO TRAINING RESULTS

YOLO training logs show you collect:

- Precision
- Recall
- mAP50
- mAP50-95

These metrics are the standard for object detection evaluation and are appropriate for reporting.

Final Epoch Metrics:

epoch	metrics/precision(B)	metrics/recall(B)	metrics/mAP50(B)	metrics/mAP50-95(B)
8	0.9685	0.9584	0.9884	0.9414
9	0.9669	0.9704	0.9898	0.9615
10	0.9752	0.9573	0.9911	0.9634
11	0.9688	0.9713	0.9903	0.9564
12	0.9688	0.9666	0.9897	0.9612

FIGURE 3. Final YOLO gender-detection metrics per epoch (Precision, Recall, mAP@0.50, mAP@0.50–0.95)

1) Label Statistics Visualizations

- Shows spatial distribution of bounding boxes over all images.
- Helps confirm balanced dataset and consistent face detection.

Labels_correlogram.jpg:

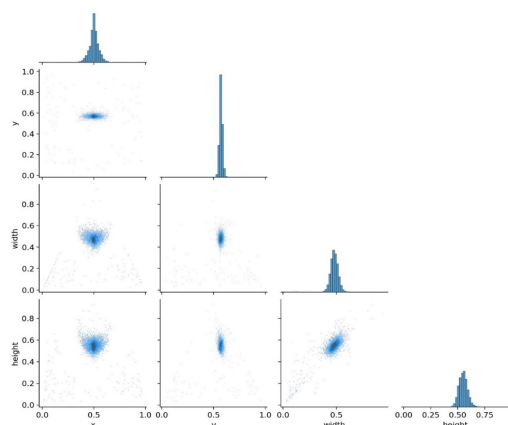


FIGURE 4. Bounding-box parameter distributions and correlations (x, y, width, height) for the gender dataset annotations

Results.p

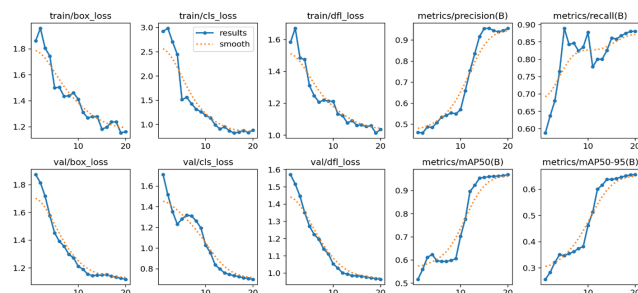


FIGURE 5. YOLO training results summary plot showing loss trends and detection metrics over epochs

BoxF1 Curve

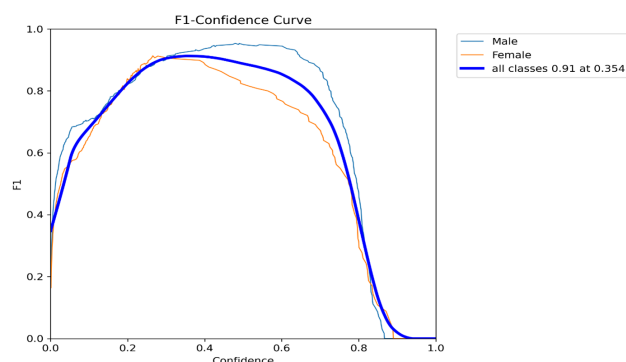


FIGURE 6. F1-confidence curve for the YOLO gender detector, showing class-wise (Male/Female) and overall performance, with the best overall F1 0.91 at confidence threshold 0.354

BoxP Curve

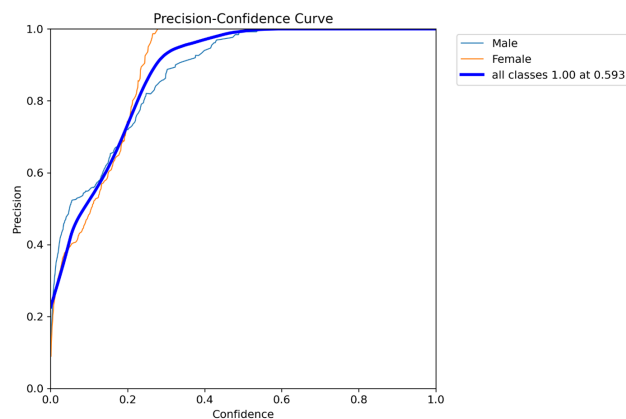


FIGURE 7. Precision-confidence curve for the YOLO gender detector (Male/Female and overall), showing overall precision reaching 1.00 at a confidence threshold of 0.593

BoxR Curve

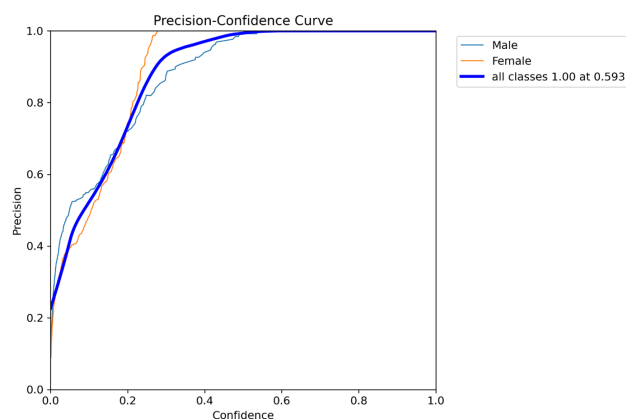


FIGURE 8. Recall-confidence curve for the YOLO gender detector (Male/Female and overall), showing recall 0.99 at confidence 0.00 and a gradual recall drop as the threshold increases, reflecting the expected trade-off between keeping detections and avoiding low-confidence predictions

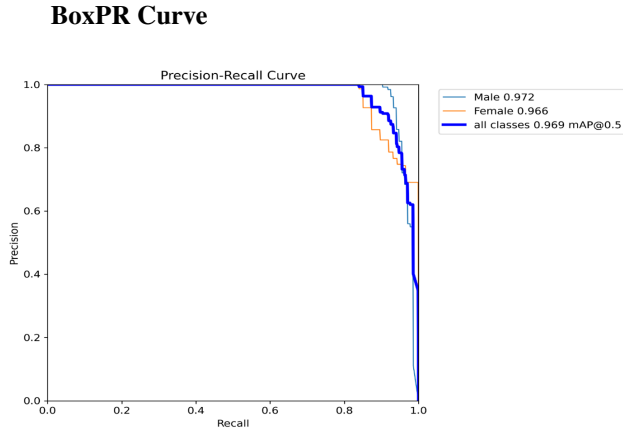


FIGURE 9. Precision–Recall curve for the YOLO gender detector, showing class-wise performance (Male/Female) and overall mAP@0.5 0.969.

Confusion Matrix

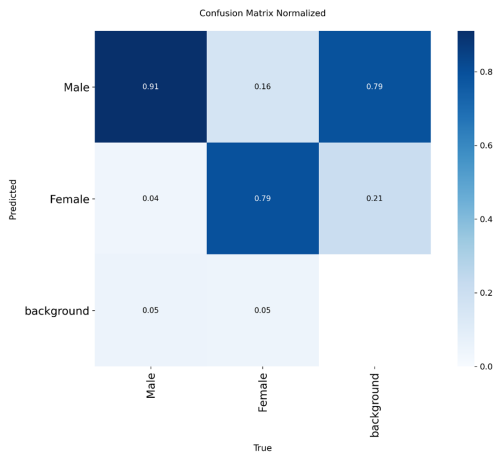


FIGURE 10. Normalized confusion matrix for the YOLO gender detector, showing classification performance across Male, Female, and background classes

VI. FUTURE DIRECTIONS

Future work can extend this system in several practical and research-driven directions:

- 1) **Increase Celebrity Classes:** Expand from 15 identities to 50+ identities and evaluate scalability under larger class counts.
- 2) **Move from Classification to Embedding-Based Recognition:** For large identity sets, embedding-based recognition (ArcFace-style) can scale better than softmax classification.
- 3) **Improve Unknown Handling:** Introduce confidence thresholding, temperature scaling, and open-set recognition strategies to reduce false positives on unseen identities.
- 4) **Real-Time Video Support:** Extend the pipeline to video by processing frames sequentially and applying face tracking to avoid repeated predictions for the same person.

- 5) **Bias and Fairness Checks:** Evaluate performance differences due to lighting, makeup, camera quality, pose, and demographic imbalance to improve robustness and fairness.
- 6) **Deployment:** Convert models to TensorRT/TFLite and deploy as a:
 - web application (Streamlit/Flask),
 - mobile prototype, or
 - desktop tool.

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